

Differential abundance analysis

0.1 Differential abundance analysis

Analysis information:

- Taxonomic level: genus_prevalent

Taxa that are less prevalent than this (in relabundance) have been agglomerated:

- Detection threshold: 0.1%
- Prevalence threshold: 0.0714286
- Number of taxonomic groups: 139

BMI is not significantly different between conditions or time points. Hence we can ignore it in the differential abundance testing. In group “B” it seems to be borderline significant ($p < 0.05$ before multiple testing adjustment).

Number of significant findings within time points.

- Kruskal test, Benjamini-Hochberg FDR adjustment

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before  after
      0      2
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Number of significant findings within treatments:

- Paired Wilcoxon test, Benjamini-Hochberg FDR adjustment

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rice  oat
    25  10
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Let us look more closely to the significant hits. The logFC corresponds to logFC in CLR (tp2-tp1).

Group	Taxon	logFC.clr	FDR	mean.relundance1	mean.relundance2
rice	Candidatus_Cibionibacter	-0.0147	0.0002	2.336	0.822
oat	GGB4569	0.0006	0.0071	0.029	0.090
rice	Clostridiaceae_unclassified	0.0082	0.0383	0.925	1.771
oat	GGB27106	0.0010	0.0405	0.019	0.115
oat	Candidatus_Cibionibacter	-0.0052	0.0680	1.548	1.014
rice	Firmicutes_unclassified	0.0025	0.0854	0.379	0.631
oat	Butyricicoccus	0.0002	0.0863	0.034	0.051
oat	Parasutterella	0.0006	0.1436	0.076	0.135
oat	Turcibacter	0.0008	0.1445	0.058	0.133
oat	Lawsonibacter	-0.0005	0.1445	0.164	0.111
rice	GGB34797	0.0049	0.1639	0.355	0.851
rice	GGB79734	0.0006	0.1639	0.116	0.178
rice	Bilophila	0.0003	0.1639	0.074	0.101
rice	GGB9559	0.0002	0.1639	0.035	0.056
rice	GGB27106	-0.0004	0.1639	0.094	0.050
rice	Anaerostipes	-0.0055	0.1639	2.504	1.931
rice	Lachnospiraceae_unclassified	-0.0123	0.1639	3.707	2.400
rice	Bacteroides	0.0095	0.1648	3.975	5.003
rice	GGB2982	0.0005	0.1648	0.098	0.145
rice	GGB14025	0.0002	0.1648	0.040	0.064

Group	Taxon	logFC.clr	FDR	mean.relabundance1	mean.relabundance2
rice	Bifidobacterium	-0.0171	0.1648	6.265	4.463
rice	Lacrimispora	0.0042	0.1796	0.762	1.198
rice	GGB9760	0.0008	0.1796	0.199	0.277
rice	Gordonibacter	0.0003	0.1796	0.049	0.082
oat	Dialister	0.0036	0.1922	1.051	1.416
oat	GGB6612	0.0000	0.1922	0.031	0.031
oat	Eubacteriales_unclassified	-0.0026	0.1922	0.786	0.525
rice	Eubacteriales_unclassified	0.0017	0.1924	0.546	0.723
rice	GGB51647	-0.0015	0.1924	0.215	0.060
rice	Mogibacterium	0.0002	0.2085	0.032	0.048
rice	Anaerobutyricum	0.0023	0.2243	1.239	1.472
rice	Enterocloster	-0.0008	0.2243	0.139	0.063
rice	GGB9342	-0.0013	0.2243	0.244	0.111
rice	Prevotella	-0.0097	0.2243	1.994	0.941
rice	Eubacteriaceae_unclassified	0.0005	0.2245	0.069	0.115

Volcano Plot: Effect Size vs Significance



